NATIONAL LEVEL AI HACKATHON

DOMAIN :- CYBERSECURITY

PROBLEM STATEMENT TITLE :-

DEEPFAKE & IMAGE MANIPULATION DETECTION FOR PREVENTING REFUND FRAUD ON DIGITAL COMMERCE PLATFORMS

TEAM NAME :- return0

IDEA TITLE & PROPOSED SOLUTION

PROBLEM STATEMENT

- Delivery platforms (Zomato, Swiggy, Blinkit, Amazon) approve refunds based solely on user-uploaded images
- Fraudsters use AI tools to manipulate expiry dates, labels, or generate synthetic complaint images
- Returns fraud cost merchants **USD 103 Billion globally (2023) — 45% of Indian consumers affected**
- No automated system exists to verify the authenticity of complaint product images

PROPOSED SOLUTION — VeriSight

- AI-powered image verification system that auto-analyzes every complaint image
- **4-layer parallel detection: CNN Forensics → ViT AI Detection → GAN Analysis → OCR Verification**
- Outputs an Authenticity Confidence Score (0–100) in < 5 seconds
- Fully explainable decisions — heatmaps + SHAP reports for human reviewer audit

HOW IT ADDRESSES THE PROBLEM

- Platform integrates VeriSight API in complaint workflow
- Image analyzed in real-time → score returned in < 5 sec
- Auto-approve (85–100) / Fast-track (60–84) / Flag (0–59)

INNOVATION & UNIQUENESS

- First system to specifically target expiry date manipulation in delivery fraud
- 4-layer multi-modal AI — no single point of failure
- OCR + forensics combined — novel for refund fraud detection

TECHNICAL APPROACH

TECHNOLOGIES & TOOLS

AIML

- EFFICIENTNET-B4 (CNN)
- VIT-B/16 (TRANSFORMER)
- CLIP-RESNET50 (GAN)
- EASYOCR + PADDLEOCR

FRONTEND

- REACT.JS DASHBOARD
- DOCKER + KUBERNETES
- GITHUB ACTIONS CI/CD
- AWS / GCP DEPLOY

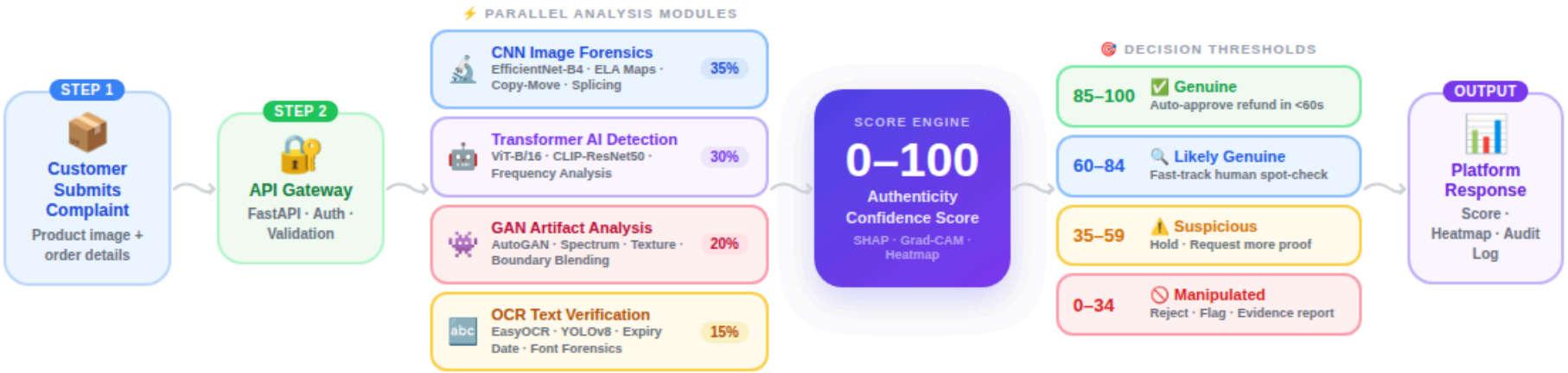
BACKEND

- FASTAPI (ASYNC REST)
- CELERY + REDIS (QUEUE)
- POSTGRESQL (DB)
- ONNXRUNTIME (INFERENCE)

XAI & CV

- SHAP (FEATURE ATTRIBUTION)
- GRAD-CAM + GCA-NET
- YOLOV8 (TEXT DETECTION)
- OPENCV + PILLOW

SYSTEM ARCHITECTURE & FLOW



FEASIBILITY & VIABILITY

FEASIBILITY ANALYSIS

- All models available as open-source pre-trained weights (PyTorch Hub, HuggingFace)
- CASIA 2.0, DF2023, ImagiNet — public benchmark datasets ready for training
- FastAPI + Celery stack enables sub-5-sec parallel inference — proven in production systems
- REST API design: platform integration in < 1 day for Zomato, Swiggy, Blinkit, etc.

CHALLENGES & RISKS

- Adversarial attacks: fraudsters may adapt to detection patterns over time
- OCR accuracy on poor lighting / damaged packaging (degraded images)
- New generative AI models emerging faster than detection benchmarks
- False positives: genuine customers incorrectly flagged if threshold miscalibrated

CONSTRAINTS

- No platform-specific labeled product complaint dataset exists publicly
- Model inference requires GPU for sub-5-sec latency at scale
- Score thresholds need platform-specific calibration to minimize false positives

MITIGATION STRATEGIES

- Custom dataset built from 4,000+ labeled product images across 5 manipulation classes
- ONNX Runtime + model quantization for fast CPU inference during early deployment
- Human reviewer override for 35–84 score range — XAI report enables transparent auditing

IMPACT & BENEFITS

TARGET USERS / STAKEHOLDERS

- Delivery platforms: Zomato, Swiggy, Blinkit, BigBasket, Amazon, Flipkart
- Platform fraud & trust-safety teams
- Genuine consumers — protected from inflated prices due to fraud losses
- Delivery partners & sellers — protected from false damage claims

SOCIAL / ECONOMIC IMPACT

- Saves platforms millions in fraudulent refund payouts annually
- Reduces human reviewer workload — auto-approve 85%+ of genuine claims
- Faster refunds for genuine users (60 sec vs 24–48 hrs manually)
- Combats organized refund fraud rings like the Rs 5.5 Cr Meesho case

LONG-TERM BENEFITS

- Model continuously improves as new fraud patterns are detected and labeled
- Extensible to insurance claims, warranty claims, B2B commerce fraud
- SHAP reports usable as legally defensible digital evidence in fraud cases
- Builds consumer trust: genuine customers get faster, fairer resolutions

RESEARCH & REFERENCES

DATASETS USED

- CASIA 2.0 — 12,614 image forgery dataset (splicing, copy-move)
- DF2023 — 1M+ images for robust generalization
- ImagiNet — 200,000 real vs AI-generated images (Boychev et al.)
- Custom dataset — 4,000+ labeled product complaint images (built from scratch)

TOOLS, APIs & REFERENCES

- HuggingFace Transformers — ViT-B/16, CLIP-ResNet50 pre-trained models
- OpenAI CLIP — GAN fingerprint detection (CVPR 2023, 99.4% accuracy)
- EasyOCR + PaddleOCR — product packaging text extraction
- Loop Returns 2024 | McAfee Global Survey 2024 | IEEE 2024

KEY RESEARCH PAPERS

- Image Forgery Detection via ELA-CNN, VGG19, EfficientNet-B2
- GCA-Net: Gated Context Attention for Image Forgery Localization
- LGrad: Learning on Gradients for Generalized GAN Image Detection
- AutoGAN: Detecting GAN-Fake Images via Spectrum Analysis
- Advanced Detection of AI-Generated Images via Vision Transformers
- Expiry Date Recognition with Improved DBNet + CBAM
- Integration of LLM in Expiration Date Scanning (R-CNN: 91.1% precision)
- Generalized FCN Framework for Expiration Date Recognition